

Homework 1

Problem 1

In this problem, we explore another example of catastrophic cancellation. Check Eq. (13) of the following paper and lines 213-216 in `tron.cpp` from the software package `LIBLINEAR` version 2.11. Explain how we avoid catastrophic cancellations.

- Paper: <http://www.csie.ntu.edu.tw/~cjlin/papers/logistic.pdf>
- `LIBLINEAR` version 2.11: <http://www.csie.ntu.edu.tw/~cjlin/liblinear/oldfiles>

Note: We do not consider the latest version of `LIBLINEAR` because this code segment has been removed.

Problem 2

Let us see another cancellation example. On page 5, line 12 of the paper

<http://www.csie.ntu.edu.tw/~cjlin/papers/plattprob.pdf>,

there are two methods to calculate $1 - p_i$, and the results produced by the two methods may differ under some conditions. Please write a short C++ program to reproduce the examples and discuss what you find. Include your code in the report.